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New paleomagnetic results of Paleocene volcanic rocks from the Lhasa block: Tectonic implications for the collision of India and Asia

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ABSTRACT

A combined geochronological and paleomagnetic investigation has been performed on Paleocene volcanic sequences in the Lhasa block near the localities of Mendui (30.1°N/90.9°E). A total of 15 sites have been sampled from rhyolitic tuffs. Stepwise thermal demagnetizations successfully isolated high unblocking temperature characteristic directions. The tilt-corrected mean direction is $D/I = 359.0°/26.1°$ with $\alpha_{95} = 9.2°$ and $N = 14$ sites, corresponding to a paleopole at 73.6°N, 274.3°E with $A_{95} = 7.3°$. Positive fold tests suggest a primary origin for the characteristic remanence. In order to provide a more accurate pole, we propose to combine site-mean directions from this study and Achache et al.'s (1984). The combined average palaeomagnetic direction from early Paleocene volcanic rocks is $D = 356.6°$, $I = 25.9°$, $\kappa = 21.7$, $\alpha_{95} = 6.8°$ after tilt correction, $N = 22$ sites, corresponding to a pole at 73.2°N, 282.4°E with $A_{95} = 5.4°$. The paleomagnetic results yield a paleolatitude of 13.6 ± 5.4°N for the southern margin of Eurasia at ~55 Ma. Compared with expected paleomagnetic directions from the stable India and Eurasia blocks, significant crustal shortening of 1400 ± 600 km and 2000 ± 550 km respectively may have occurred between the southern margin of Eurasia and the stable India, and within Eurasia since the collision of India and Eurasia.
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1. Introduction

The collision between the India and Eurasian continents is considered to be the major tectonic events of Cenozoic time. The precise timing of the initial India–Asia contact and the amount of crustal shortening remains a matter of dispute until recently. Based on different methods, the proposed initial time of collision, ranges from 34 Ma to 70 Ma (Searle et al., 1987; Rage et al., 1995; Rowley, 1998; Yin and Harrison, 2000; Ding, 2003; Mo et al., 2007; Aitchison et al., 2007). Paleomagnetic data are among the strongest qualitative constraints that can be sought to reconstruct the block boundaries and their history. Some suggest that the onset of collision of India with Asia is inferred to be at 55 Ma from paleomagnetic determination of an abrupt slowdown in the northward velocity of the India plate (Patriat and Achache, 1984; Besse and Courtillot, 1988; Klootwijk et al., 1992). At that time, paleomagnetic data indicate that rocks of northern Greater India (the northernmost part of greater India prior to the collision of India and Eurasia) experienced a remagnetization event attributed to be the initial India–Asia contact at equatorial to low northern latitudes (Klootwijk et al., 1981; Tong et al. 2008). However, the collision of India with an intra-oceanic island during the Paleocene may also provide an explanation for the slowdown in convergence between India and Asia at 55 Ma (Aitchison et al., 2007). Clearly, the paloposition of the southernmost Eurasia during Cenozoic time is of prime importance for understanding the collision of India and Eurasia and the kinematics. Previous paleomagnetic studies on the early Cenozoic in the Lhasa block have played a key role in reconstructions of the southern margin of Eurasia block (Achache et al., 1984; Lin and Watts, 1988; Otofuji et al., 1991; Chen et al., 1993). In fact, hard quantitative data on the early Cenozoic position of the Lhasa block are limited. Achache et al. (1984) reported the Paleocene paleomagnetic results of the Lizizong Formation from 8 sites and 46 samples, and proposed a paleolatitude of 13.5 ± 6.5°N for the Lhasa area. These values imply crustal shortening by as much as 1900 ± 850 km north of the Lhasa block and by 1400 ± 1200 km south of the block since the onset of India–Eurasia contact. Lin and Watts (1988) presented two sets of paleomagnetic results from Paleocene–Eocene red beds near Erdaogou on the Qiangtang (North Tibet) block, however, these data from the Fenghoushan Formation yielded inconsistent results, with respective paleolatitude of 7 ± 4°N and 18 ± 3°N. These paleomagnetic results appear to be partly remagnetized since some of their primary directions are very close to the present geomagnetic field (PGF) in geographic coordinates, and provided no fold or reversal tests. Otofuji et al. (1991) focused on the 42 Ma Quxu pluton and The Lizizong Formation. Only two sites yielded primary

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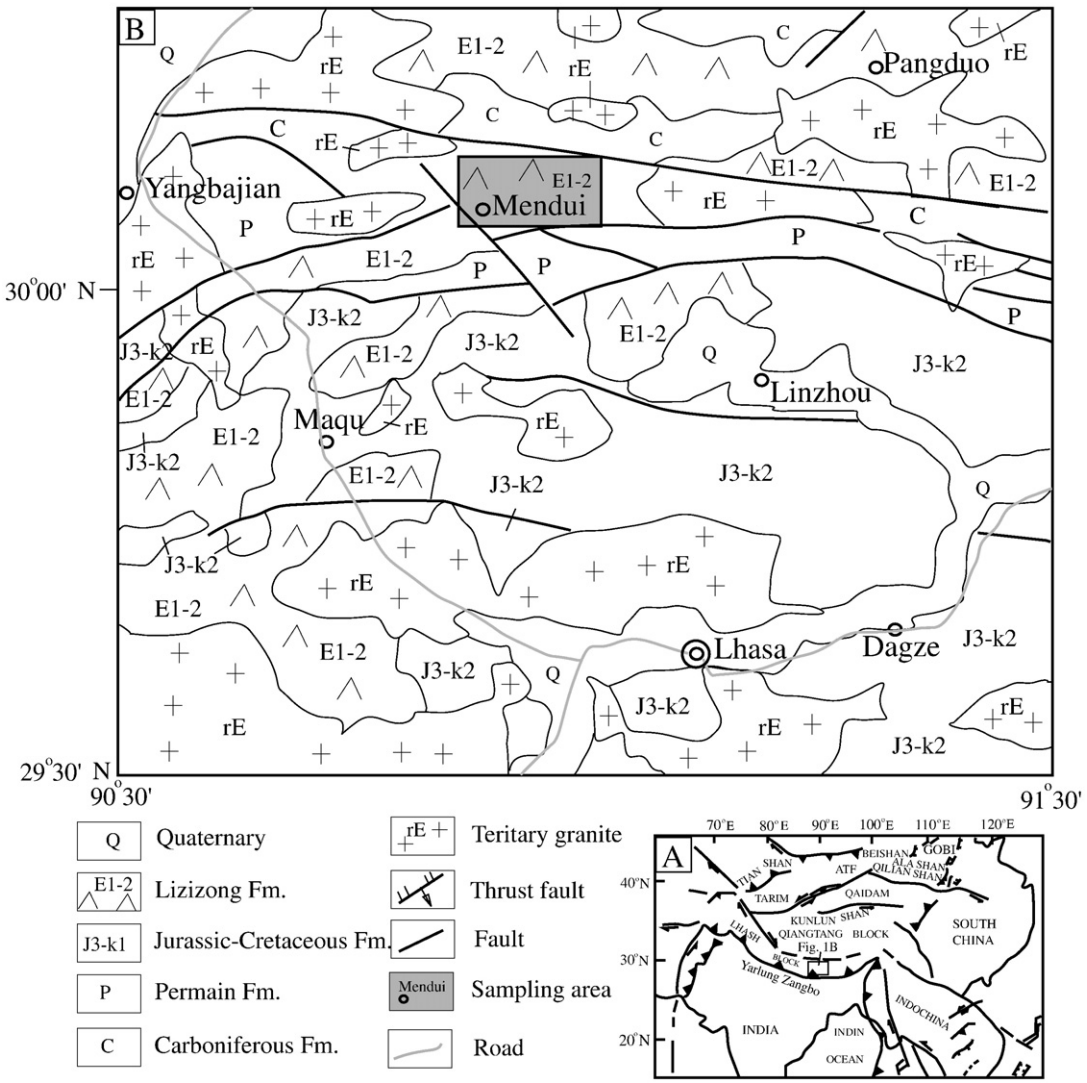


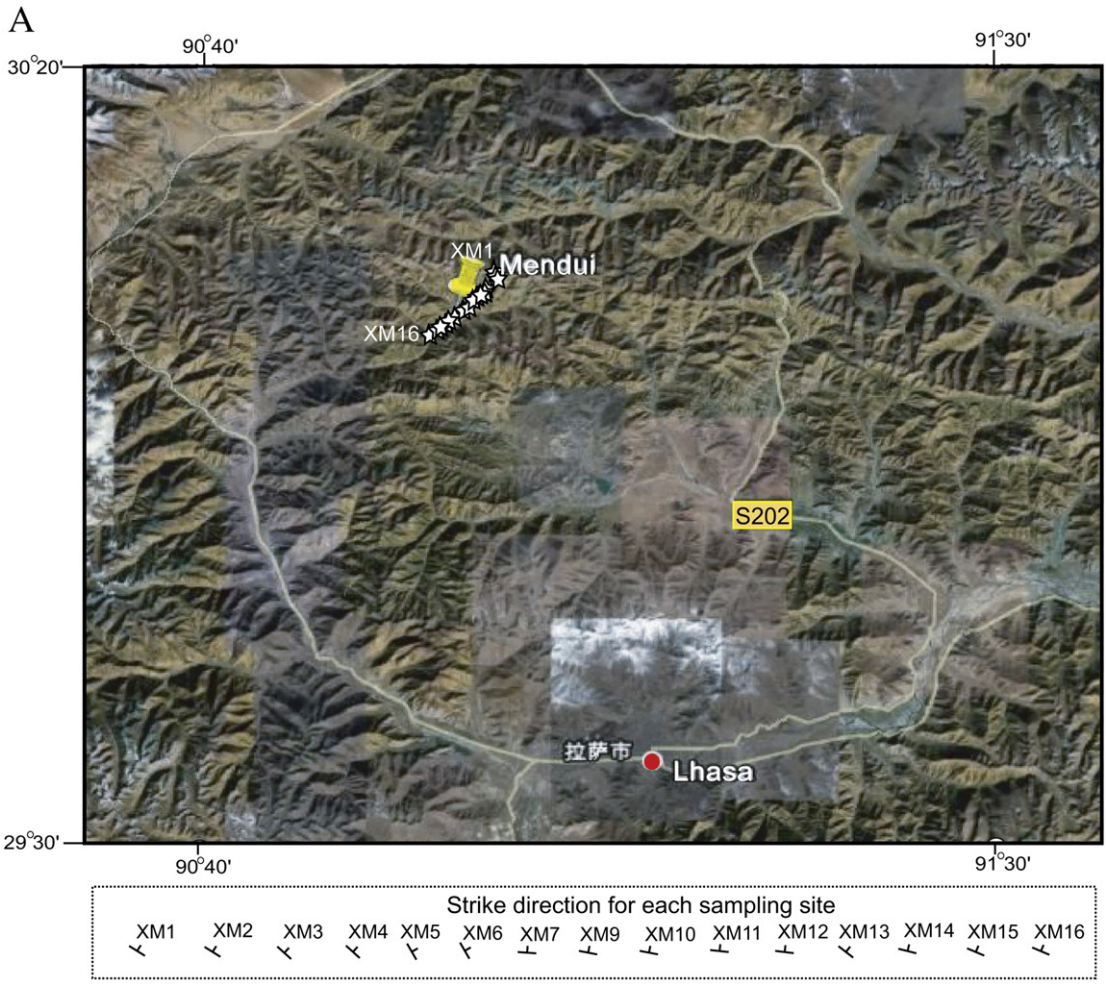
Fig. 1. (A) Simplified tectonic map of Central Asia centered on Tibet, showing blocks and major faults. (B) Simplified geological map of the investigation area.

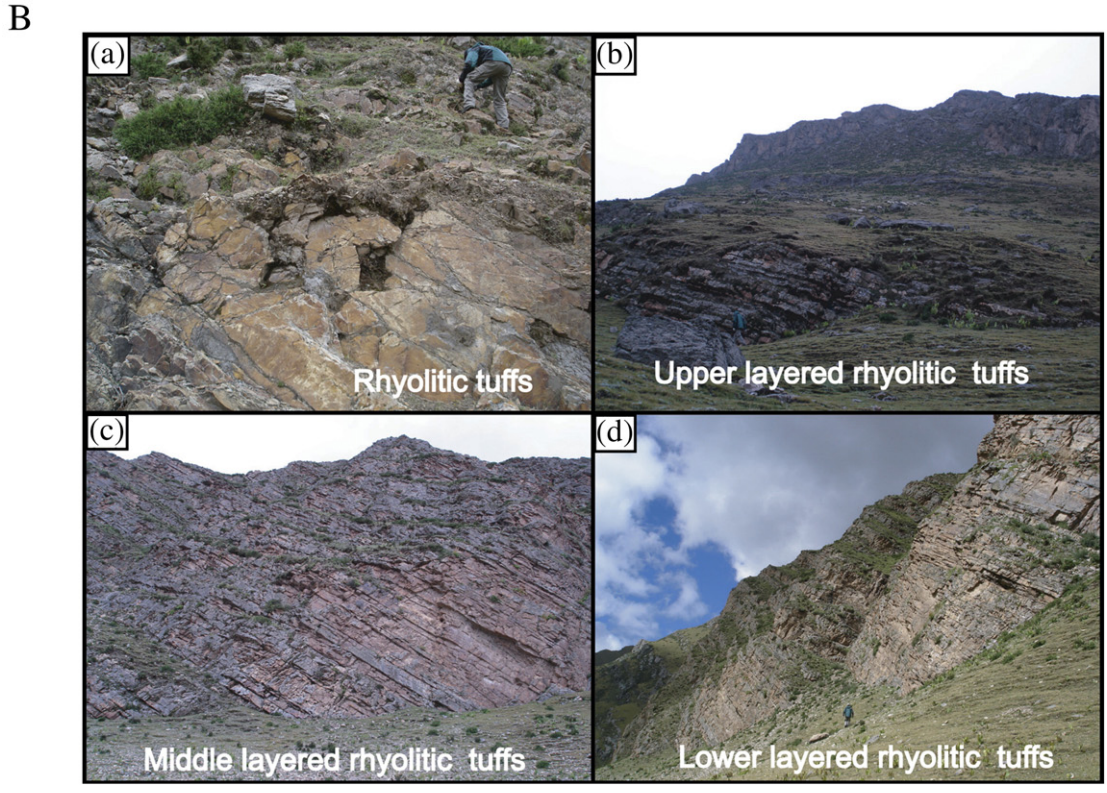
directions with $\alpha_{95}<20°$ for the Quxu pluton, and one site gave a paleomagnetic direction for the Lizizong Formation. As with many studies of granites, the results were far from perfect due to the coarse texture associates with such rocks and rather low magnetic intensities. Chen et al. (1993) sampled Tertiary red beds in the western Lhasa block. In fact, there are no fossils to confirm their age, and the paleomagnetic directions show a very large scatter, both within site as well as between sites due to tectonic or remagnetization problems (Chen et al., 1993). The review of previous work shows that existing paleomagnetic database is still scarce for the early Cenozoic rocks of the Lhasa block. In order to improve the paleolatitude of the southern margin of Eurasia, new paleomagnetic and geochronology data from the early Cenozoic volcanic sequences in the Lhasa block are presented here to provide the necessary paleolatitude control on the southern margin of Eurasia block during early Cenozoic and constraint on the crustal shortening due to collision of the India–Eurasia.

2. Geological setting and sampling

The Tibetan plateau consists essentially of four blocks between India to the south and the Tarim block to the north (Fig. 1A) (Chang and Cheng, 1973). They include the Lhasa block, the Qiangtang block, and the Kunlun and the Qaidam blocks in the NE. The Yarlung–Zangbo suture separates India from the Lhasa block and is mostly an early Cenozoic suture associated with the India–Asia collision (Tapponnier et al., 2001). The amount of crustal shortening between Greater India and Eurasia due to the onset of India–Asia contact remains a subject of debate. Plate tectonic reconstructions based on paleomagnetic data from the Himalayan orogenic belt suggest that 2600 ± 900 km of post-collision convergence has taken place between Eurasia and Greater India, with 1700 ± 610 km of this total accommodated by north–south shortening in the Tibetan plateau and lateral tectonic escape (Patriat and Achache, 1984; Achache et al., 1984; Besse and Courtillot, 1991). Published estimates of shortening in Himalaya based on paleomagnetic data range ~700 km and 1500 km (Patzelt et al., 1996). The Himalayan shortening (south of Indus suture zone) is estimated by mass balances cross-section at ~670 km from Pakistan to Sikkim (DeCelles et al., 2002). However, by balanced cross-section, Guillot et al. (2003) estimated a total Himalayan shortening of about 1100 km since 55 Ma. The difference between these average values is explained by an extending of Greater India up to 650–700 km and the early subduction of the India plate at the Indus suture zone.

Fig. 2. (A) Topographic map of the studied area showing details of paleomagnetic sampling locations, site distributions, and strike direction and dip azimuths for each sampling site. The sampling sites are marked by five stars. (B) Photographs showing the Nianbo Formation strata in the studied section. (a) Rhyolitic tuff. (b) The upper rhyolitic tuff sequence. (c) The middle rhyolitic tuff sequence. (d) The lower rhyolitic tuff sequence.


A
90°40'
91°30'
30°20'
29°30'
XM1
Mendui
XM16
S202
拉萨市
Lhasa
Strike direction for each sampling site
XM1 XM2 XM3 XM4 XM5 XM6 XM7 XM9 XM10 XM11 XM12 XM13 XM14 XM15 XM16


B
(a)
Rhyolitic tuffs
(b)
Upper layered rhyolitic tuffs
(c)
Middle layered rhyolitic tuffs
(d)
Lower layered rhyolitic tuffs

The Lhasa block is the southernmost Eurasia block, between the Yarlung–Zangbo suture to the south and the Bangong suture to the north. The Lizizong volcanic rocks (~65–45 Ma) are widely distributed in the Lhasa block along Yarlung–Zangbo suture. On the basis of the detailed mapping and eruption hiatuses, these volcanic sequences can be subdivided into lower Dianzhong (64.4–60.6 Ma), middle Nianbo (~54 Ma) and upper Pana Formations (Mo et al., 2007). $^{40}Ar/^{39}Ar$ dating on andesite and plagioclase from the Dianzhong Fm volcanic rocks gives a plateau age ranging from 64.5 to 60.6 Ma (Maluski et al., 1982). Zhou et al. (2004) reported a $^{40}Ar/^{39}Ar$ plateau age of 56.5 Ma for the Nianbo olivine trachyandesite in the mid-parts of the Formation. $^{40}Ar/^{39}Ar$ dating gives an age range of 50 to 40.8 Ma for the Pana Fm rhyolitic rocks (Mo et al., 2003; Zhou et al., 2004). These Formations have a maximum thickness of ~5 km and overly unconformably the Upper Cretaceous and older marine and continental sedimentary strata, and are overlain with angular unconformity by the Oligocene Rigongla Formation (K–Ar: 31.4 Ma) (BGMRX, 1993; Mo, et al., 2007), indicating the age of folding of the Lizizong volcanics is prior to the Oligocene.

The Nianbo Formation has a thickness of approximately ~1400 m in the sampled section. Fifteen sites from the Nianbo Formation, which mainly consists of layered rhyolitic tuffs intercalated with tuffaceous sandstone and mudstone, were sampled at Mendui area (30°5.054′N/90°55.97′E) located to ~60 km to the northwest of Lhasa City (Figs. 1B and 2A,B). Oriented hand samples were taken because it is logistically difficult to collect paleomagnetic core samples from the rhyolitic tuffs using a portable gasoline drilling machines at height of 5000 m. Specimens were cored from the sample block in the laboratory. Hand samples were oriented with a magnetic compass. Sun compass orientations were not measured because of cloudy weather. We used the International Geomagnetic Reference Field IGRF (2007) declination of 0.1° in the Mendui region to correct for magnetic deviation. Each site comprised five to eleven blocks distributed over stratigraphic distances ranging up to 5–10 m. Each sample within the sites was collected from different layered rhyolitic tuff (Fig. 2A, B) and the studied sections display volcanic rocks, which show significantly different color and structures. The outcrop of Nianbo Formation has homoclinal dips. The bedding in the studied section dips at some 15–35° towards the N. We therefore, assume that each sample within one site may represent one cooling units. In addition, one fresh rhyolitic tuff sample was collected at the middle of the studied section for dating the Nianbo Formation volcanic sequences by a conventional Ar–Ar method.

3. Age determination and paleomagnetic results

To determine the age of Nianbo Formation, we completed $^{40}Ar/^{39}Ar$ analyses on a single grain of Plagioclase. Analytical data are listed in Table 1 and release spectra are shown in Fig. 3. Individual grain was purified using a Frantz magnetic separator and conventional heavy organic liquid separation techniques, and then selected under a binocular microscope. The separates were analyzed by $^{40}Ar/^{39}Ar$ dating using the MM-5400 micromass-spectrometer in China University of Geosciences (Beijing). The duration of irradiation and the neutron dose were 9.5 h and 2.08×10^{17} n/cm^2, respectively. Other detailed

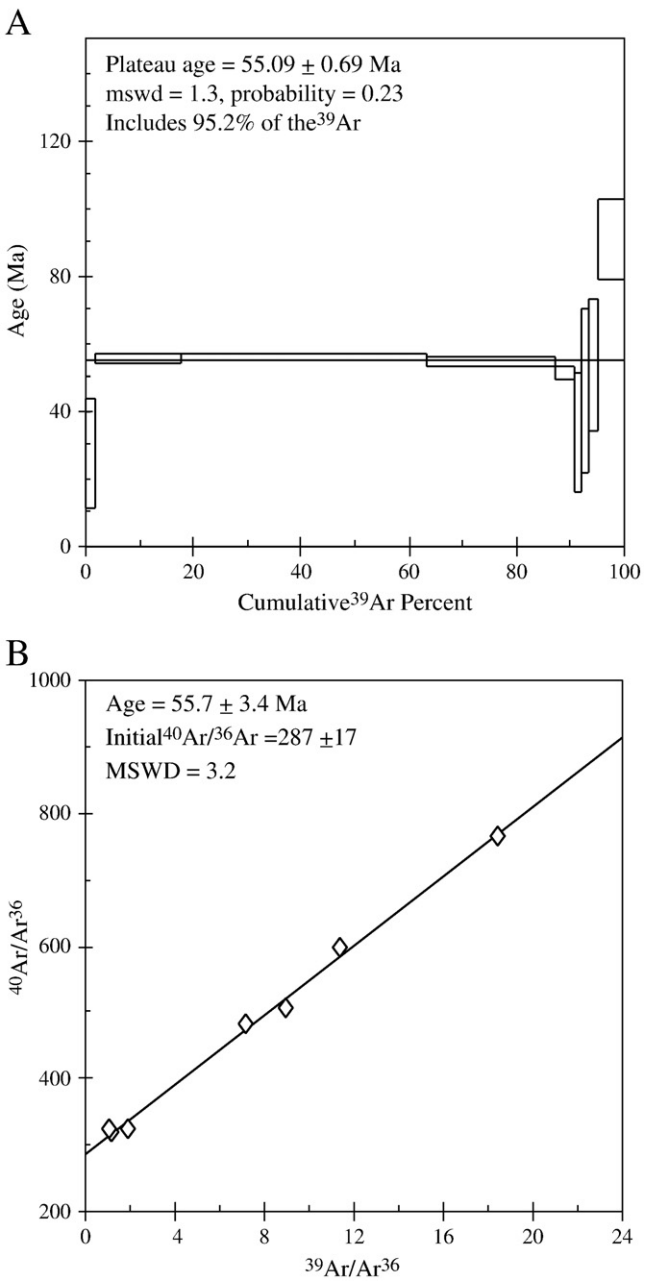


Fig. 3. Plagioclase $^{40}Ar/^{39}Ar$ plateau (A) and isochron ages (B) from the Nianbo Formation volcanic rocks.

Table 1
Dating results of $^{40}Ar/^{39}Ar$ plateau ages for plagioclase.

Temp (°C)	$^{40}Ar/^{39}Ar$	$^{36}Ar/^{39}Ar$	$^{37}Ar/^{39}Ar$	$^{40}Ar/^{39}Ar_k$	$^{39}Ar\times10^{-8}$ccSTP	^{39}Ar (%)	Age (Ma)	Error (Ma)
DB-56Pl (plagioclase), J=0.001201, Wt=0.0864 g								
550	109.0910	0.337504	42.82166	12.8313	0.028	1.61	27.59	15.95
700	67.0112	0.139284	2.57873	26.0849	0.277	16.02	55.65	1.45
800	52.1288	0.087988	2.01419	26.3088	0.789	45.59	56.12	1.01
870	40.7603	0.059621	26.86570	25.5954	0.413	23.84	54.62	1.57
960	56.7473	0.113367	7.76479	23.9426	0.059	3.43	51.14	2.11
1060	166.4150	0.512936	12.25862	15.8632	0.024	1.40	34.05	17.65
1160	265.0982	0.836426	41.93552	21.6325	0.022	1.28	46.27	24.39
1260	301.6286	0.938664	10.25911	25.1839	0.035	2.02	53.75	19.23
1500	361.4089	1.079343	4.23575	42.9049	0.083	4.81	90.64	12.08

measuring procedures have been described by Wang et al. (2007). Weighted mean plateau ages (WMPA) are included where >50% of the ^{39}Ar released in contiguous steps is within 1σ error. The single grain of Plagioclase sample yields a ^{40}Ar/^{39}Ar weighted mean plateau age of 55.09 ± 0.69 Ma (95.2% cumulative ^{39}Ar). The isochron age is 55.7 ± 3.4 Ma (initial ^{40}Ar/^{36}Ar 287 ± 17) (Table 1, Fig. 3). Thus, the plagioclase sample gives well-defined plateaus, and the isochron ages are similar to the plateau ages. Zhou et al. (2004) similarly reported a ^{40}Ar/^{39}Ar plateau age of 56.5 Ma for the Nianbo Formation in the mid-parts of the Formation. However, further Ar–Ar dating is needed to re-confirm the Ar–Ar result from this single grain of plagioclase.

All samples underwent stepwise thermal demagnetization using 14–19 steps in an ASC TD-48 oven with an internal residual field less than 10nT. Remanent magnetizations were measured with a 2G cryogenic magnetometer in the palaeomagnetic laboratory of the Institute of Geomechanics, CAGS (Beijing). Magnetization directions were determined by principal component analysis. The site-mean of paleomagnetic directions were calculated using Fisherian statistics. The anisotropy of magnetic susceptibility (AMS) was measured to assess if these rocks have suffered substantive tectonic strain. The low-degrees of anisotropy (mean $p' < 1.03$) and the large scatter in the maximum susceptibility axis directions (Fig. 4), suggests that the Nianbo Formation has not experienced significant strain due to compaction or tectonic stress. The magnetic mineralogy of representative lithologies was studied using stepwise thermal demagnetization (Lowrie, 1990) of isothermal remanent magnetization (IRM). Thermal demagnetization of the three-component IRM used fields of 0.12 T, 0.4 T and 1.2 T. Thermal demagnetization of the soft (<0.12 T)

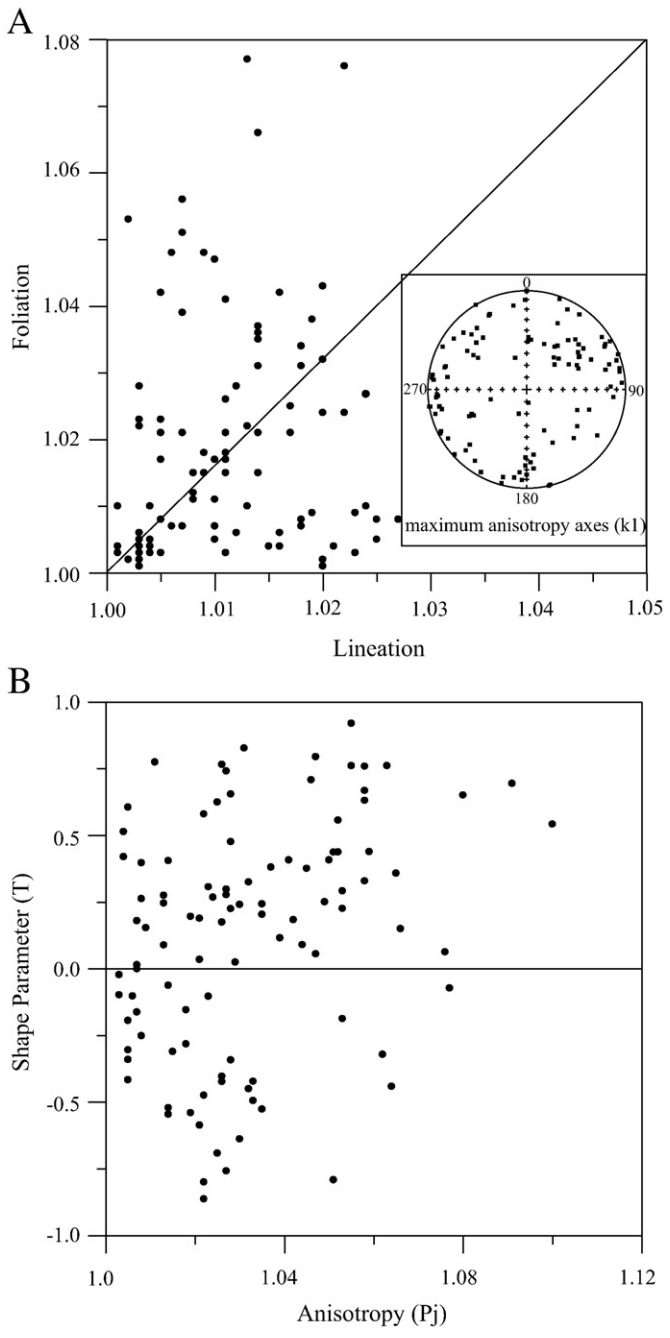


Fig. 4. (A) Anisotropy of magnetic susceptibility data in stratigraphic coordinates from the Nianbo formation volcanic rocks. Lineation = $k1/k2$, foliation = $k2/k3$. (B) Plots of shape parameter (T) versus anisotropy degree (P').

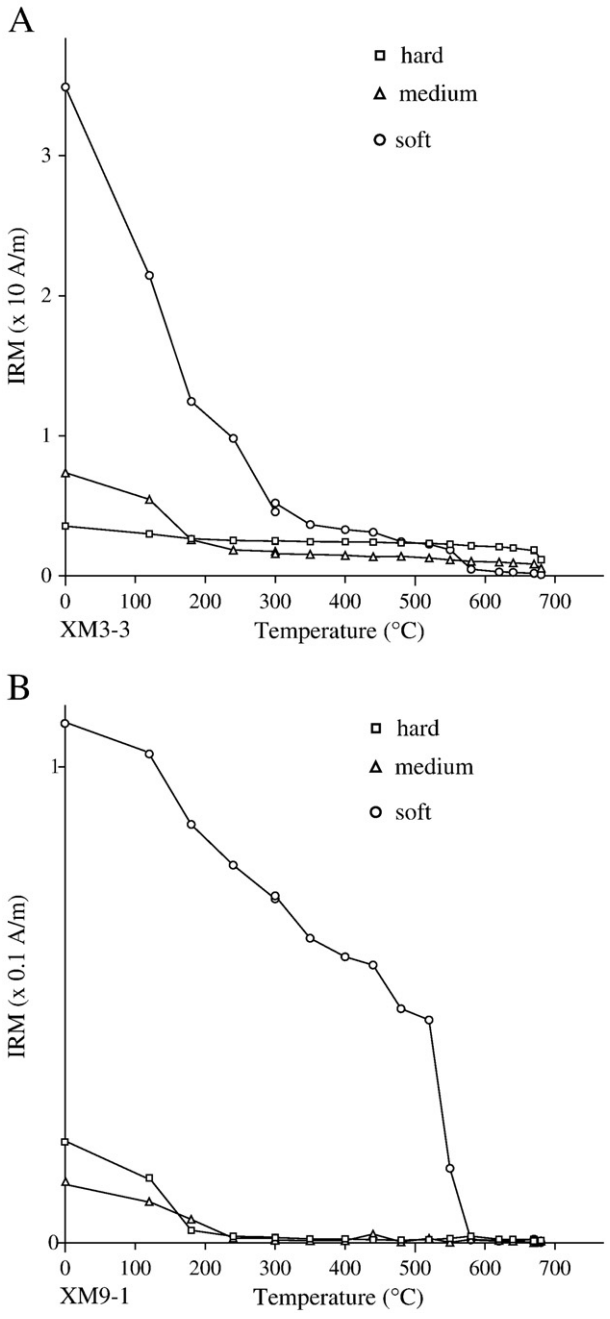


Fig. 5. (A–B) Thermal demagnetization of a three-component IRM of representative samples, showing unblocking temperatures around 580 °C or 680 °C. The soft (<0.12 T), medium (0.12–0.4), and hard (0.4–1.2 T) components are shown during thermal demagnetization.

fraction shows unblocking temperature of 580 °C, indicative of magnetite. The hard (0.4–1.2 T) and medium (0.12–0.4 T) fractions show a distinct unblocking temperature of 680 °C, which is probably indicative of hematite (Fig. 5A,B).

Intensities of natural remanent magnetization (NRM) can be subdivided into two distinct groups. In the first group (sites 1, 2, 3, 5, and 6), NRM ranges from 0.01 A/m to 0.1 A/m. The second group (sites 7, 9, 10, 11, 12, 13, 14, 15, and 16) exhibits much lower intensities,

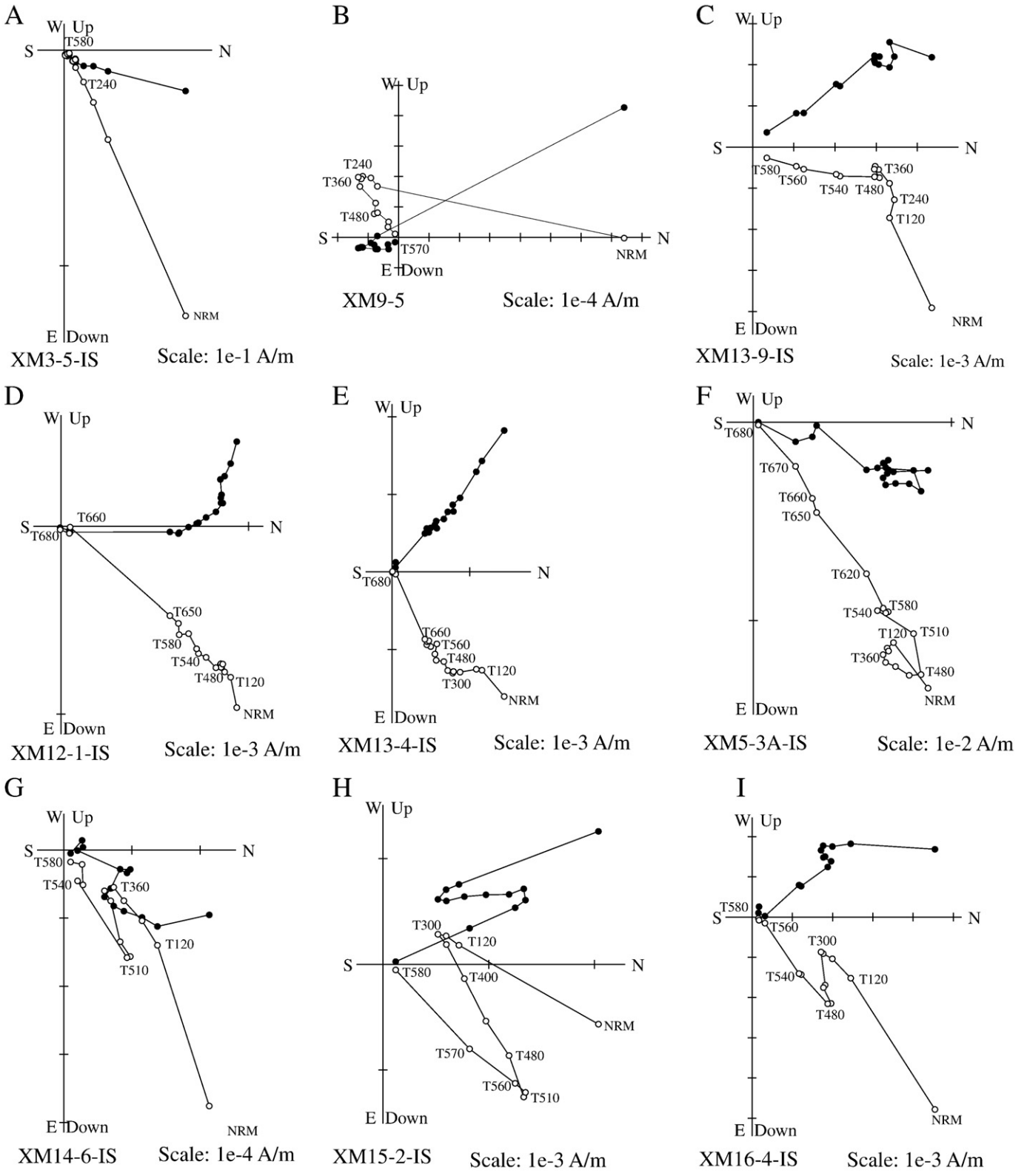


Fig. 6. Orthogonal demagnetization diagrams of representative volcanic rocks (sample number and corresponding sites listed) in situ coordinates. Demagnetization steps in °C in all plots. NRM is the natural remanent magnetization. (A–E) Typical examples from sites 3, 9, 12, 13 showing two magnetization components; (F–I) Typical examples from sites 5, 14, 15, 16 showing a small nearly antiparallel component (300–480°) superimposed on the high-temperature component.

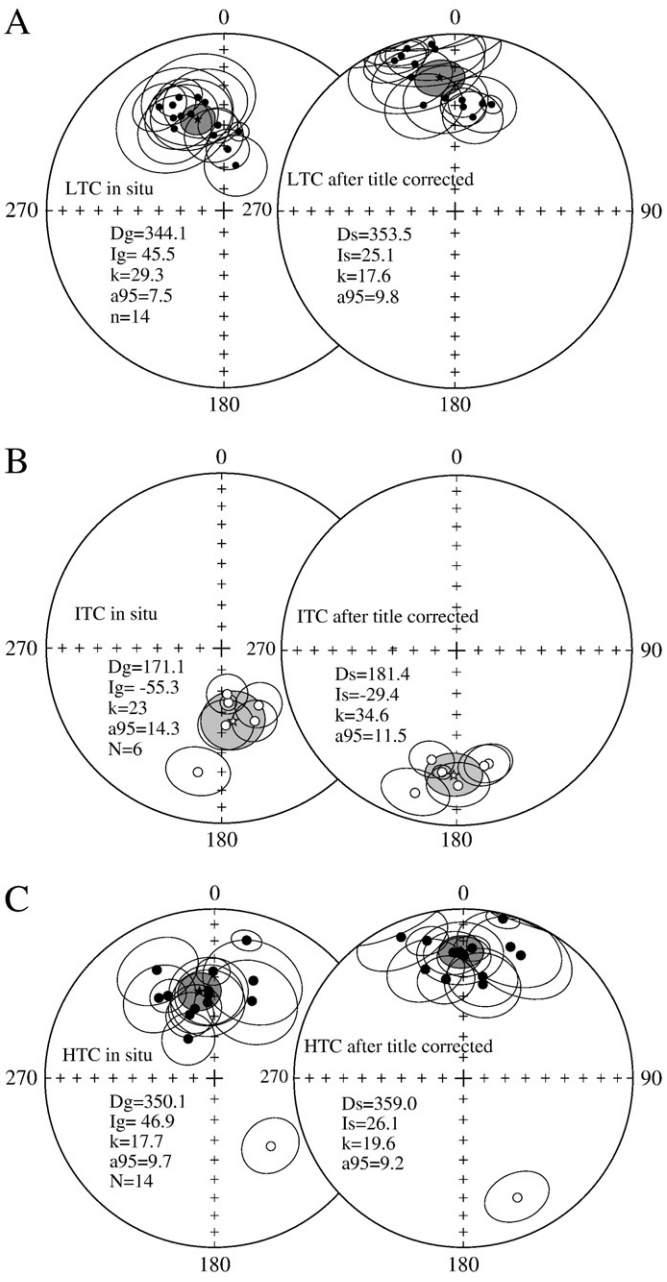


Fig. 7. (A) Equal-area projections of the low-temperature component. (B) Equal-area projections of intermediate-temperature component. (C) Equal-area projections of the high-temperature component.

ranging from 0.001 A/m to 0.0001 A/m. In general, thermal demagnetization isolated three magnetization components in most samples: (1) component A: a low-temperature component (LTC) is isolated in most of the samples below about 300 °C (Fig. 6A–I). The mean direction of this component is $D=344.1°$, $I=45.5°$ ($N=14$ and $\alpha_{95}=7.5°$ before tilt correction), and is indistinguishable from the present-day field direction ($D=0°$ and $I=46.0°$). A negative fold test for this LTC (Fig. 7A) indicates the LTC is post-fold remagnetization acquired in recent times. (2) Component B: an intermediate-temperature component (ITC) with unblocking temperatures in the range of 300–480° is isolated at sites 5, 7 11 and 14–16. The mean of the ITC ($D=181.4°$, $I=-29.4°$, $N=6$ and $\alpha_{95}=11.5°$ after tilt correction) seems to be antiparallel magnetization superimposed in the HTC (Fig. 6F–I), and yields an inconclusive fold test at 95% confidence level (Fig. 7B, Table 2). As shown in Table 2, all samples show a single reversed polarity and display relatively consistent directions. This may indicate that component B is a remagnetization acquired during reversed polarity interval after folding. (3) Component C:, a high-temperature component (HTC) is mostly present between about 510 and 580 °C or sometimes up to 680 °C (Fig. 6). The tilt-corrected mean direction of the HTC is $D_s=359.0°$, $I_s=26.1°$, $\kappa_s=19.6$, $\alpha_{95}=9.2°$, $N=14$ sites, corresponding to a pole at 73.6°N, 274.3°E with $A_{95}=7.3°$ (Table 3, Fig. 7C).

The various rhyolitic tuff layers are arranged in a simple homoclinal sequence and each sample within one site was collected from different layered rhyolitic tuff. Some sites (sites 6, 11, 12, and 13) exhibit dual polarity magnetization within the site (Table 3). Thus, we assume that each sample is a separate cooling unit and each site comprises at least several cooling units. Consequently the mean of the 14 sites from the studied section almost certainly provides adequate averaging of the geomagnetic field. The virtual geomagnetic pole (VGP) scatter of 18.3° for the sampling area falls near the limit (17.6–24.9°) of predicted VGP (McFadden et al. 1991) for 30–40° latitude and the 45–80 Ma interval. This suggests that the overall mean direction of the HTC has adequately averaged any geomagnetic secular variation.

In general, the differences in bedding attitudes between the different sites are small and thus it is difficult to obtain a meaningful fold test. However, the site-mean directions do yield a positive fold test at 95% confidence level (McFadden, 1990) (Table 3). Also, the progressive unfolding process reveals that the site-mean directions give a maximal concentration of direction at about 73.4% unfolding (with uncertainties ranging from 38.6 to 108.2%) ($D=359.8°$, $I=25.0°$, $a_{95}=9.3°$; Watson and Enkin, 1993), which is statistically indistinguishable from that of 100% unfolding ($D=359.0°$, $I=26.1°$, $a_{95}=9.2°$). Thus we consider that the HTC of Nianbo Formation at 100% unfolding indicates a prefolding origin for the ChRM and so represents the primary remanence.

Table 2
Site-mean paleomagnetic directions of the intermediate-temperature from Paleocene Volcanic Rocks in Mendui area, Northwestern Lhasa.

Site	$N(n)$	R/N	D_g	I_g	D_s	I_s	κ	α_{95}	Strike	Dip
XM5	4/10	4/0	190.8	−28.7	196.3	−16.3	51	13	330	18
XM7	4/12	4/0	176.9	−53.6	179.4	−23.8	56.8	12.3	274	30
XM11	6/8	6/0	155.6	−52.2	164.3	−33.2	45.9	10	276	21
XM14	7/11	7/0	147.1	−58	166.6	−32.7	32.6	10.7	284	31
XM15	4/5	4/0	172.8	−64.4	186.8	−30.6	597.8	3.8	293	34
XM16	7/8	7/0	173.1	−68.4	192.9	−36.1	42.6	9.4	293	35
Mean	6		171.1	−55.3	–	–	23	14.3	–	–
					181.4	−29.4	34.6	11.5	–	–

N/n: number of samples measured and used to calculate mean; R/N, samples showing reversed/normal polarity; D_g, I_g, D_s, I_s: declination and inclination in geographic and stratigraphic coordinates respectively; κ: the best estimate of the precision parameter; α95: the radius that the mean direction lies within 95% confidence. Strike: dip direction of beds − 90°, dip: the mean dip and dip azimuth of site beds; unit: degree.
(1) fold test for rocks is inconclusive according to McFadden (1990): critic Xi at 95% = 0.29, Xi1 IS = 0.91, Xi1 TS = 0.74.

Table 3
Site-mean paleomagnetic directions of the high-temperature from Paleocene volcanic rocks in Mendui area (30.1°N, 90.9°E), Northwestern Lhasa.

Site	Slat	Slong	$N(n)$	R/N	D_g	I_g	D_s	I_s	κ	α_{95}	Strike	Dip
This study												
XM1	30°5.504′	90°55.970′	9/9	0/9	359.3	37	3.8	24	51.2	7.3	302	15
XM2	30°5.242′	90°55.800′	5/7	0/5	13.1	17.5	13.9	3.3	164.4	6	302	15
XM3	30°5.250′	90°55.800′	7/7	0/7	21.7	38	25	21	8.5	22	312	18
XM4	30°4.891′	90°55.511′	No isolated HTC									
XM5	30°4.811′	90°55.477′	10/10	0/10	354.9	52.9	11.7	42.8	9.1	16.9	330	18
XM6	30°4.797′	90°55.398	7/7	1/6	356.2	48.8	10.7	38.8	16.3	15.4	330	18
XM7	30°4.755′	90°55.363′	4/12	0/4	25.9	48	19.7	18.7	19.8	21.2	274	30
XM9	30°4.513′	90°55.028′	8/9	8/0	140.4	−46.7	155.7	−23.6	17.7	13.5	281	31
XM10	30°4.448′	90°54.981′	5/6	0/5	326.3	67.2	350	40.4	37.8	12.6	281	31
XM11	30°4.430′	90°54.971′	6/8	2/4	356.1	47	358.5	26.3	15.2	17.7	276	21
XM12	30°4.306′	90°54.776′	8/9	2/6	332	28.4	336.2	9.8	10.9	17.6	277	22
XM13	30°4.306′	90°54.776′	6/7	1/5	325.3	41.9	340.7	32.4	17	16.7	308	22
XM14	30°4.279′	90°54.718′	11/11	0/11	344.3	54.9	355.4	26.4	14.9	12.2	284	31
XM15	30°4.253′	90°54.681′	5/5	0/5	330.6	43.4	345.2	17.1	85	8.3	293	34
XM16	30°4.231′	90°54.677′	8/8	0/8	338.7	56.9	0.3	28.7	25.7	11.1	293	35
Mean (1)			14		350.1	46.9	–	–	17.7	9.7		
					–	–	359.0	26.1	19.6	9.2		
Pole Long. = 274.3°E Lat. = 73.6°N, A_{95} = 7.3° Paleolat. = 13.8 ± 7.3°N												
Previous study: Achache et al., 1984												
24	30	91	6/14		151.7	−20.2	160.5	−30.9	25.1	13.6		
25	30	91	4/5		6.8	5.0	13.7	34.0	109.3	8.8		
26	30	91	7/13		207.1	−67.4	182.0	−28.3	105.8	5.9		
28	30	91	7/14		186.4	−23.0	180.0	−18.6	91.4	6.3		
39	30	91	5/5		341.4	44.6	346.4	15.7	13.6	21.5		
40	30	91	6/6		357.2	60.8	354.4	35.9	252.0	4.2		
41	30	91	6/7		167.8	−45.2	168.4	−20.9	152.8	5.4		
47	30	91	5/5		145.1	15.4	145.1	−14.5	64.7	9.6		
Mean (2)			8		170.1	−33.0	170.9	−25.5	26.4	11.0		
Mean (1) + (2)			22		350.1	42.1			10.7	9.9		
							356.6	25.9	21.7	6.8		
Pole Long. = 282.4°E Lat. = 73.2°N, A_{95} = 5.4° Paleolat. = 13.6 ± 5.4°N												

Slat:latitude of site; slong: longitude of site; and same symbols as in Table 2.
(1) Fold test for this study is positive according to McFadden method (1990): critic Xi at 95% = 4.44, Xi2 IS = 4.74, and Xi2 TS = 0.87.
(2) Reversals test for this study and Achache et al. (1984) is positive: angle between the mean directions $r = 14.1 < r_{critic} = 14.6°$, classification C. (McFadden and M.W.McElhinny, 1990).

4. Results – discussion

4.1. Previous results from Lhasa block

From the results of Ar–Ar dating and palaeomagnetic analysis given above, we propose a new Paleocene palaeopole for the Lhasa block (Table 4, Fig. 8), located at λ = 73.6°N, φ = 274.3°E (A_{95} = 7.3°), indicating a palaeolatitude of 13.8 ± 7.3°N for the studied area (reference point: 30°5.504′N/90°55.97′E) located about 100 km the north of the Yarlung–Zangbo suture. Therefore, the Paleocene paleolatitude of the studied areas may represent the paleolatitude of the southern margin of Eurasia at ~55 Ma. The other available early

Table 4
Locality paleomagnetic poles from this study compared to expected directions from adjacent blocks (reference point: 30.1°N, 90.9°E).

Block or areas	Age (Ma)	Lat.	Lon.	$N(n)$	Plat.	Plon.	A_{95} (dp/dm)	Comment	Reference
Lhasa Block	45–65	30	91	8(46)	71.4	299.8	11	No FT	Achache et al., 1984
	45–65	29.8	90.8	1(4)	67.9	311.8	10.5	No FT	Otofuji et al. 1991
	~55	30.1	90.9	14(104)	73.6	274.3	7.3	PFT	This study
	~55	30.1	90.9	22(150)	73.2	282.4	5.4	PFT	This study + Achache et al., 1984
Qiangtang Block	Pa–E	34.6	92.4	15	61	253	3.8/7.4	No FT	Lin and Watts, 1988
	Pa–E	34.6	92.8	12	55	201	3.2/5.6	No FT	Lin and Watts, 1988
Tarim	60	40.28	75.38	32	54.8	167.8	6.4	PFT	Huang et al. 2005
	54	40.7	76.1	18	77	187.5	4.4	PFT	Bazhenov et al. 2002
Gamba	55–63	28	89.2	14	65.4	277.6	7.5	PFT	Patzelt et al. 1996
India APWP	55				60.7	277.7	4.8		Besse and courtillot, 2002
Europe APWP	55				81.4	168.3	4.2		Besse and courtillot, 2002

Block	Age (Ma)	Lat.	Lon.	A_{95}	Relative to Tarim		Relative to Eurasia		Relative to Gamba		Relative to India	
					Rotation	Flattening	Rotation	Flattening	Rotation	Flattening	Rotation	Flattening
					$R \pm \triangle R$	$F \pm \triangle F$	$R \pm \triangle R$	$F \pm \triangle F$	$R \pm \triangle R$	$F \pm \triangle F$	$R \pm \triangle R$	$F \pm \triangle F$
Lhasa Block	~55	30.1	90.9	5.4	−18.0 ± 5.5	14.2 ± 5.1	−13.3 ± 5.5	18.0 ± 5.0	0.1 ± 5.4	−12.6 ± 5.3	0.6 ± 6.9	−8.0 ± 6.8

Lat. (Lon.): latitude (longitude) of sites; plat (Plon): latitude (longitude) of pole; $N(n)$: sites (samples); A_{95}/(dp/dm): the radius that the mean pole lies within 95% confidence; no FT: without fold test; PFT: positive fold test; Pa–E: Paleocene–Eocene; rotation $R \pm \triangle R$ is the vertical axis rotation with 95% confidence limit. Flattening $F \pm \triangle F$ is the flattening of inclination with 95% confidence limit. Rotation and flattening are derived from observed direction minus expected direction at locality (reference point: 30.1°N, 90.9°E) calculated from reference pole. −/+: southward/northward or counterclockwise (clockwise rotation); and unit: degree.

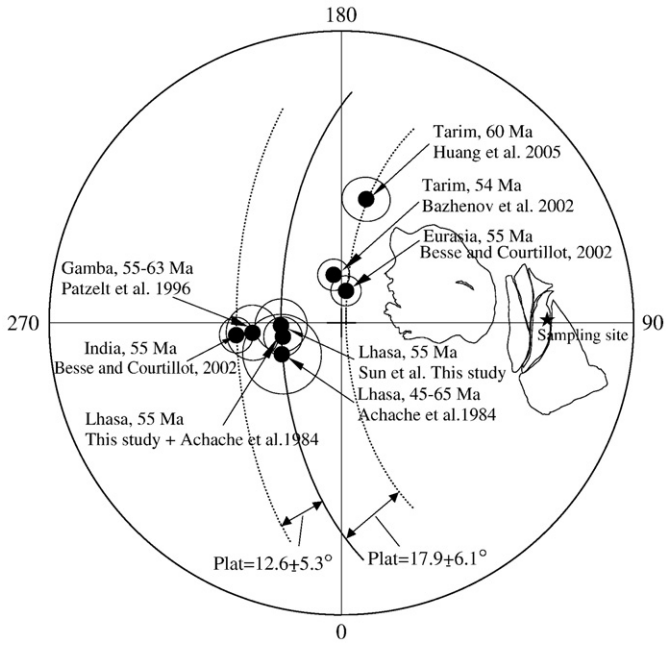


Fig. 8. Equal-area projection of Paleogene poles of the Lhasa Block and adjacent blocks. See Table 4 for values and references. Dotted line: small-circle running through the 55 Ma reference pole of India (Eurasia) centered on our site locations; solid lines: small-circles passing through our site locations; stars: sampling location.

Cenozoic data from the Lhasa block and the adjacent blocks or areas are listed in Table 4, and the corresponding paleopoles are shown in Fig. 8. Three paleomagnetic studies of early Cenozoic rocks are available within the Lhasa and Qiangtang blocks (Fig. 8 and Table 4). Achache et al. (1984) proposed a paleolatitude of 13.5 ± 6.5°N for the Lhasa block in the Paleocene–Eocene. Otofuji et al. (1991) reported a paleomagnetic result of the Lizizong Formation, although the number of sites and samples is too small to draw any significant inference (Table 4). Lin and Watts (1988) produced two early Cenozoic paleomagnetic datasets from the Qiangtang block, but these are inconsistent (and from the same area) and no field tests were used to assess the reliability of these data (Table 4). As illustrated in Fig. 8, we find that our new early Cenozoic paleopole is statistically distinguishable (at 95% confidence level) from the paleopole of the adjacent Lhasa locality of Achache et al. (1984). Because these results (this study and Achache et al.'s) are not statistically distinguishable and because they are from the similar age range (~55 Ma) and areas, we have combined the site-mean directions from both studies to provide a more robust paleomagnetic pole. The combined final average palaeomagnetic direction for early Paleocene volcanic rocks is $D = 356.6°$, $I = 25.9°$, $\kappa = 21.7$, $\alpha_{95} = 6.8°$ after tilt correction, $N = 22$ sites, corresponding to a pole at 73.2°N, 282.4°E with $A_{95} = 5.4°$. Based on the 22 distinct volcanic sites of Table 3, we obtain a VGP scatter of 17.4, which falls not far from the 17.6–24.9° range of VGP scatter predicted for 30–40° latitude during the 45–80 Ma interval (McFadden et al. 1991). This suggests that the overall mean direction of HTC from both studies (This study and Achache et al., 1984) from Lhasa areas have averaged any secular variation. Likewise the reversals tests are positive at the 95% confidence level (McFadden and McElhinny, 1990) (Table 3), indicating a prefolding origin for the characteristic magnetization. Taken together, the positive fold and reversal tests, high-temperature ChRM stability, and precise determination of sample ChRM directions suggest a primary origin for the characteristic magnetization.

4.2. Comparison to reference poles

Our paleomagnetic results from Nianbo Fm volcanic rocks in the eastern Lhasa block provide a key constraint on the paleoposition of the southern margin of Eurasia at ~55 Ma, the time of the collision. In order to reconstruct the paleoposition of the Lhasa block prior to collision with India and to estimate the magnitude of crustal shortening, calculated paleolatitudes are compared with expected paleolatitudes from stable India and Eurasia for the sampled locality at Mendui which include Achache et al.'s data (Fig. 8; Table 4). The expected paleolatitudes from the reference Indian paleopole at 55 Ma (Besse and Courtillot, 2002) and from the Gamba area (the northern extension of Great India) at 55–63 Ma (Patzelt et al., 1996) show significant relative north–south displacement (12.6° ± 5.3° and 8.0° ± 6.8° respectively), compared to the studied area, but no significant rotation (−0.1° ± 5.4° and -0.6° ± 6.9° respectively) since ~55 Ma (Fig. 8, Table 4). The north–south shortening amounts to 1400 ± 600 km when compared with stable India block and 900 ± 750 km with the Gamba area, at the longitude of the sampling sites (90.9°E) since the time of remanence acquisition (~55 Ma). This shortening may result from subduction of a limited amount of continental crust along the Zangbo suture zone and from underthrusting of continental crust along major thrust faults such as the main Boundary Thrust (MBT), main Central Thrust (MCT), and Kangmar thrust faults south of the suture zone. Several estimates of crustal shortening have been proposed for the Indian continent. It is generally accepted that crustal shortening in Himalaya exceeded 360 km (Yin and Harrison, 2000), and that the amount of India underthrusting beneath southern Tibet along the Main Central Thrust (MCT) is about 220–400 km (Zhao et al., 1993). Estimates of subsequent shortening within the central Himalaya suggest that ~500 to 1000 km of Indian continental lithosphere existed north of what is now north–central India (DeCelles et al., 2002; Royden et al., 2008). Our result of about 1400 ± 600 km crustal shortening exceeds these values. It could be argued that structural geology provides minimum estimates on crustal shortening because they do not include penetrative strain or small-scale folds and faults, which could significantly increase the total shortening.

Using the 55 Ma mean pole of the Eurasian apparent polar wander path (APWP) proposed by Besse and Courtillot (2002), the expected declination and inclination for the sampling area are 9.9° and 50.9°N, respectively (reference site: 30.1°N, 90.9°E). Comparing the reference direction deduced from Eurasian APWP with that observed from the Lhasa block, the declination and latitudinal differences with the sampling area of the Lhasa block are 13.3 ± 5.5° and 18.0 ± 5.0° at ~55 Ma respectively (Fig. 8; Table 4). These results could imply a large intracontinental crustal shortening of 2000 ± 550 Km between the Lhasa block and Eurasia (or Siberia) at the sampling longitude (90.9°E), and a significant anticlockwise rotation with respect to Eurasia since ~55 Ma. The amount of the counterclockwise rotation of the Lhasa block is remarkably similar to that in the northern extent of Greater India (Gamba) and in India (Table 4). The relative motion of India with respect to Eurasia shows the continuous counterclockwise motion during the Cenozoic (Patriat and Achache, 1984). In addition, Southern Tibet and Eurasia had already been sutured before the collision of Asia–India. Thus we attribute this counterclockwise rotation of the Lhasa block to the co-rotation of India. Further paleomagnetic data and structural study will be needed to test this concept.

The paleomagnetic results of Paleocene volcanic rocks from the Tarim block imply there is little or no paleolatitude difference between the Tarim block and the reference latitude of Eurasia at ~60 Ma (Bazhenov et al., 2002; Huang et al., 2005). Furthermore, when the Lhasa block is compared with the results of Paleocene volcanic rocks (~54 Ma) from the Tarim block, a discrepancy in paleolatitude (14.2 ± 5.1°) and paleomagnetic declination (18.0 ± 5.5°) can be inferred, which would imply a significant magnitude of 1600 ± 600 km crustal shortening between these blocks at the sampling longitude (90.9°E), as well as a anticlockwise rotation since Paleocene. After the collision of India–Asia, intracontinental convergence and deformation continued

across Tibet, extending northward to involve a large area. Such large amount of N–S convergence inferred by the paleomagnetic data have been previously accounted for by a combination of internal deformation (folding and thrusting), and large lateral displacements (extrusion) along the Altyn Tagh and Red River faults (Dewey et al., 1989; Yang and Besse, 1993; Jin et al., 1996; Yin and Harrison, 2000; Chen et al., 2002).

5. Conclusions

We have reported new paleomagnetic results from 14 sites of the early Cenozoic Nianbo Formation collected near Mendui in the Lhasa block. A high-temperature characteristic magnetic component is isolated and passes a fold test. We argue that this is a primary magnetization which can be dated at ~55 Ma. Since the corresponding direction is consistent with the results from Achache et al.'s study (1984), we prefer to combine site-mean directions from both studies to provide a new Paleocene pole based on a higher number of sites for Lhasa block. When the results are compared to the studies available from the adjacent blocks, new paleomagnetic data from the Lhasa block suggest that 1400 ± 600 km of post-collision shortening has taken place between the southern margin of Eurasia and the stable India, and some 2000 ± 550 km of shortening within Eurasia since the collision of India and Eurasia.

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